

Ques In a ΔABC , if $a^2 + b^2 = 101c^2$, then find the value of

$$\frac{\cot(C)}{\cot(A) + \cot(B)}$$

Sol

Given $a^2 + b^2 = 101 \cdot c^2$

$$\begin{aligned} \frac{\cot(C)}{\cot(A) + \cot(B)} &= \frac{\left(\frac{a^2 + b^2 - c^2}{4\Delta} \right)}{\left(\frac{b^2 + c^2 - a^2}{4\Delta} \right) + \left(\frac{a^2 + c^2 - b^2}{4\Delta} \right)} \\ &= \frac{a^2 + b^2 - c^2}{2c^2} = \frac{101c^2 - c^2}{2c^2} \\ &= \boxed{50} \text{ Ans} \end{aligned}$$

Formula

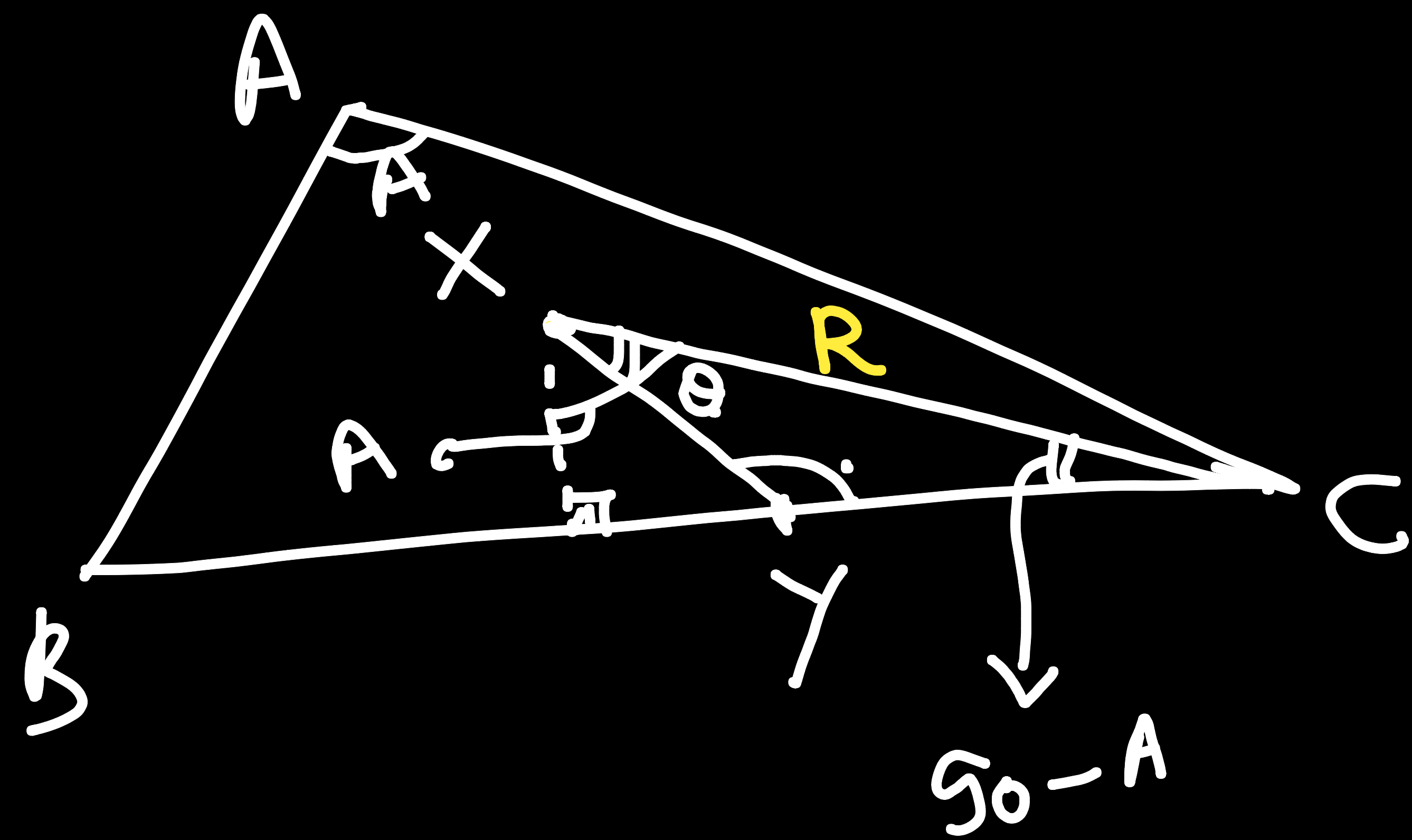
$$\cot(A) = \frac{b^2 + c^2 - a^2}{4\Delta}$$

$$= \frac{\cot(A)}{\sin(A)} = \frac{b^2 + c^2 - a^2}{2bc \sin(A)}$$

$$\Delta = \frac{1}{2} bc \sin(A)$$

Ques Let X is the circumcentre of $\triangle ABC$ and Y is a point on the side BC . Find the circumradius of $\triangle CXY$, when Y tends to coincide with 'C'.

Sch.



Let circumradius of ΔCXY is R'

$g_m \Delta x Y C$

$$\frac{XYC}{XC} = 2R' \sin(\angle XYC)$$

$$R = 2R' \sin(\pi - (0 + 90^\circ - A)) = 2R' \sin A$$

$$R = 2R' \sin(\pi - (\theta + 90^\circ - A)) = 2R' \cos(\theta - A)$$

when $Y \rightarrow C$

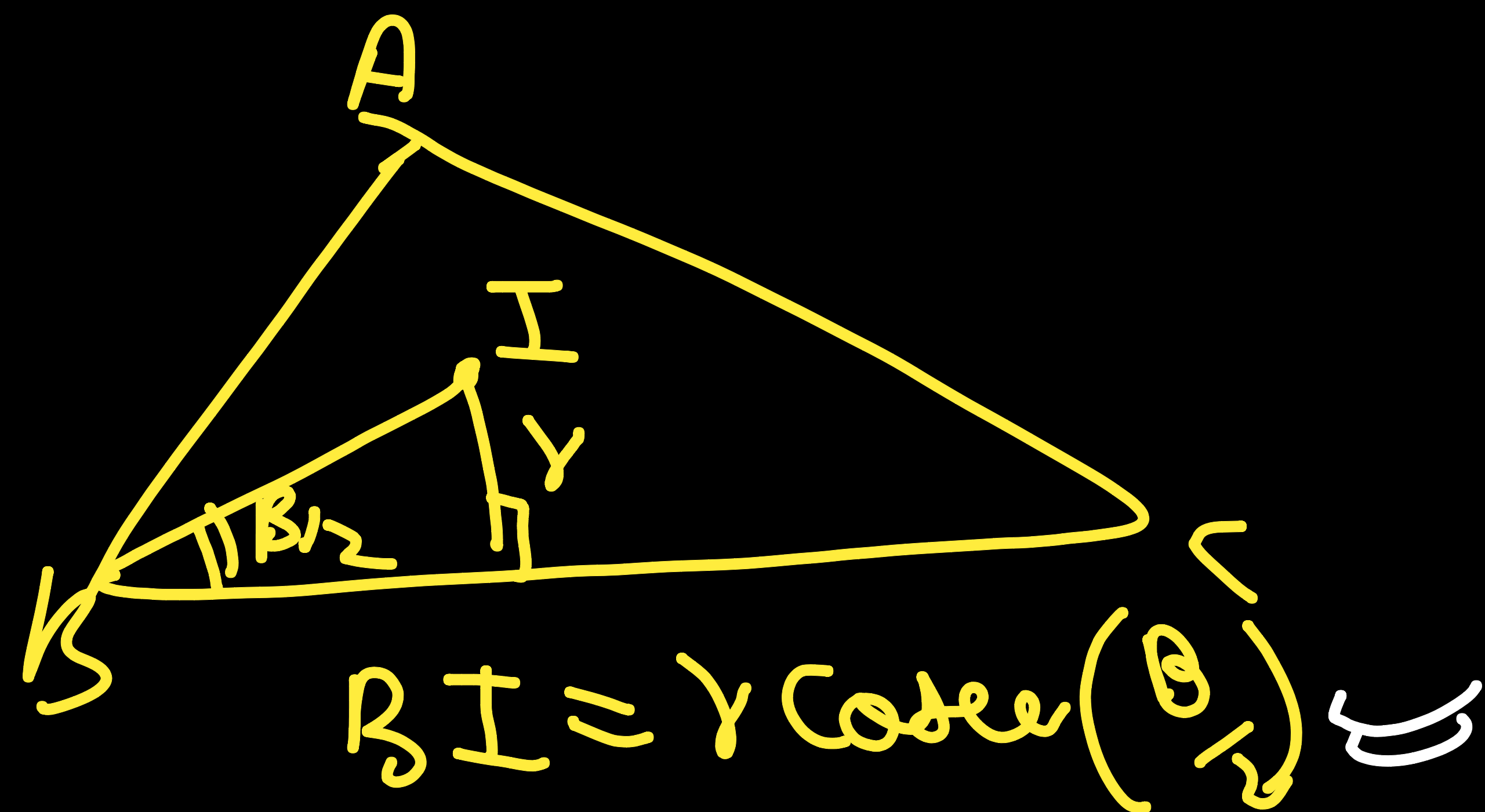
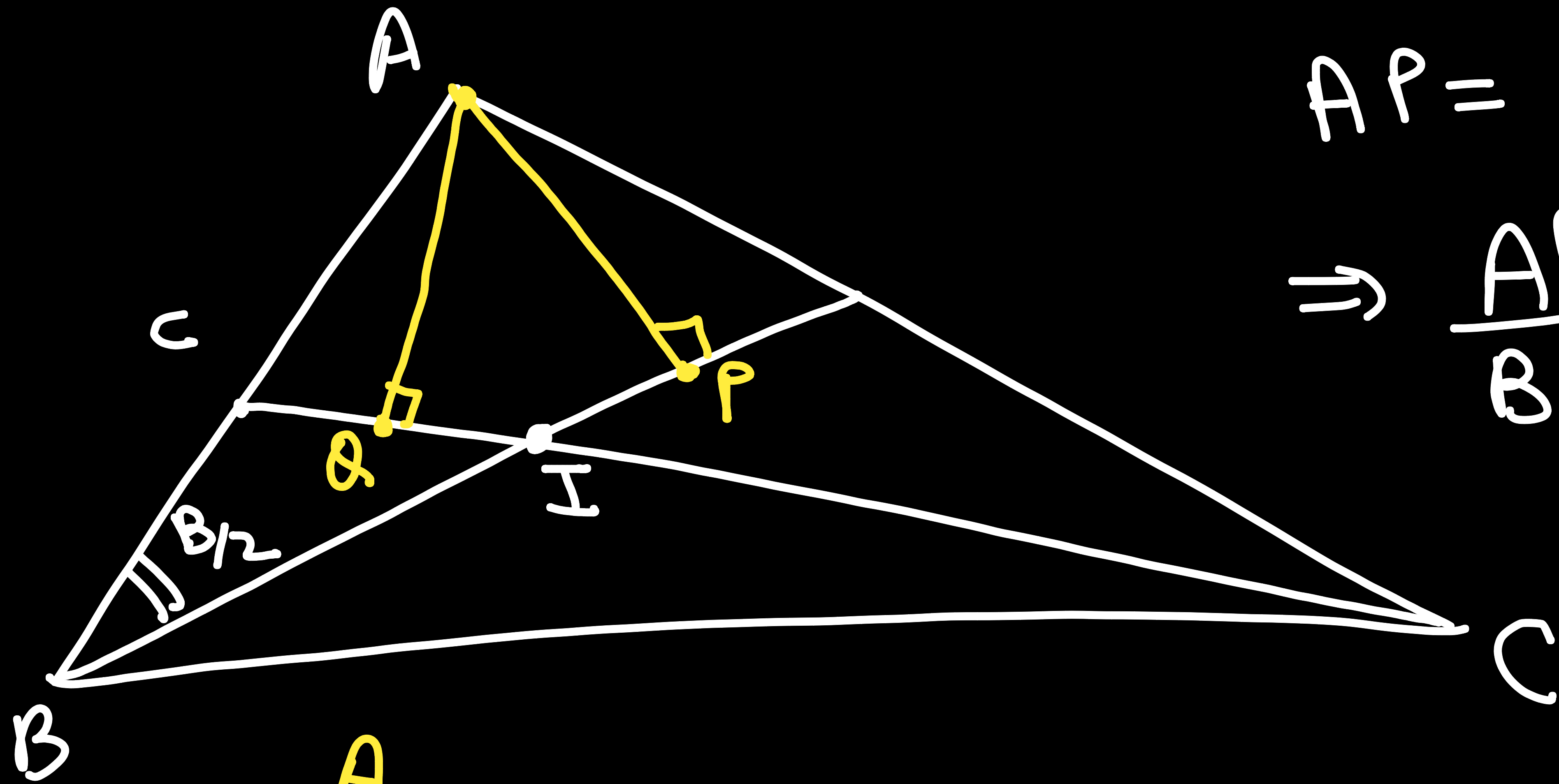
$$R' = \frac{R}{2 \cos(A - \theta)}$$

when $Y \rightarrow C$
then $\theta \rightarrow 0$

$$R' = \frac{R}{2 \cos(A)}$$

Ques Let ABC be a triangle with incentre as I and P, Q be the feet of the perpendiculars from A to BI and CI respectively, if $A = 60^\circ$ then find the value of $\frac{AP}{BI} + \frac{AQ}{CI}$.

Soln.



In $\triangle ABP$.

$$AP = AB \cdot \sin(B/2) = c \sin(B/2)$$

$$\Rightarrow \frac{AP}{BI} = \frac{c \cdot \sin(B/2)}{r \cdot \csc(B/2)} = \frac{c \cdot \sin^2(B/2)}{r} = \frac{c}{2r} \cdot (1 - \cos(B))$$

$$\frac{AP}{BI} = \frac{c - c \cos(B)}{2r}$$

$$\text{Similarly } \frac{AQ}{CI} = \frac{b - b \cos(C)}{2r}$$

$$\begin{aligned} \frac{AP}{BI} + \frac{AQ}{CI} &= \frac{b + c - (c \cos(B) + b \cos(C))}{2r} = \frac{b + c - a}{2r} \\ &= \frac{\frac{2r}{2} (b + c - a)}{2r} = \frac{s - a}{r} \end{aligned}$$

$$\begin{aligned}
 \frac{AP}{BI} + \frac{AQ}{CI} &= \frac{x-a}{y} \\
 &= \cot(A/2) \\
 &= \cot(30^\circ) \\
 &= \boxed{\sqrt{3}} \text{ Ans}
 \end{aligned}$$

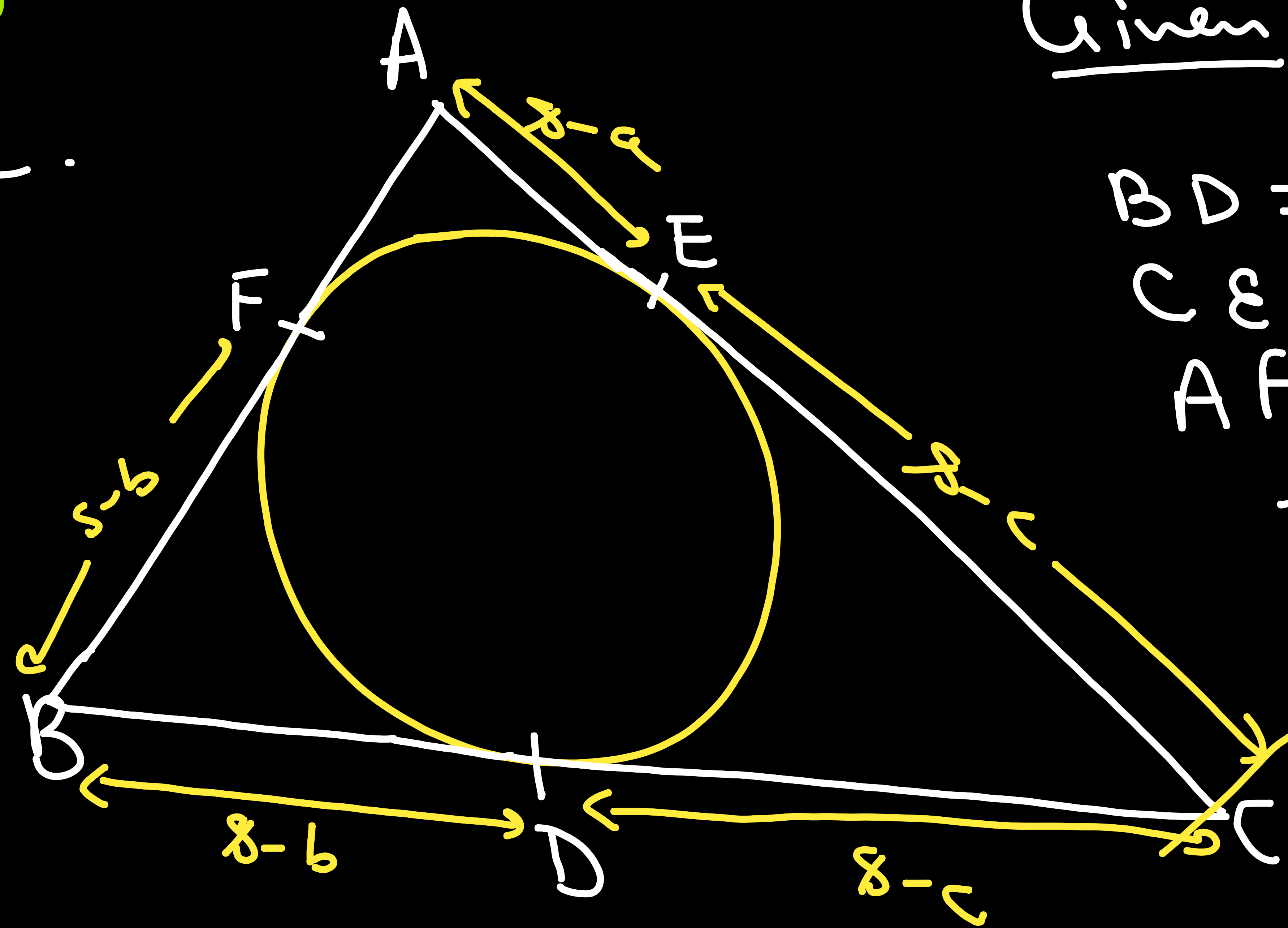
Formula

$$y = (x-a) \tan(A/2)$$

$$\Rightarrow \frac{x-a}{y} = \cot(A/2)$$

Ques In $\triangle ABC$, incircle touches side BC, CA and AB at D, E & F respectively. If the radius of incircle is 4 units and if BD, CE and AF are consecutive integers, find the length of sides of the triangle ABC.

Soln.



Given, $r = 4$.

$$BD = n = s - b$$

$$CE = n + 1 = s - c$$

$$AF = n + 2 = s - a$$

$$+ \quad 3n + 3 = 3s - 28$$

$$s = 3(n + 1)$$

$$r = 4 \Rightarrow \frac{\Delta}{s} = 4 \Rightarrow \Delta^2 = 16s^2$$

$$\Rightarrow s \cdot (s-a) \cdot (s-b) \cdot (s-c) = 16s^2$$

$$(n+2) \cdot n \cdot (n+1) = 16 \cdot 3 \cdot (n+1)$$

$$n^2 + 2n = 48$$

$$\hookrightarrow n = 6, -8$$

$$s - a = 8 \rightarrow c = 14$$

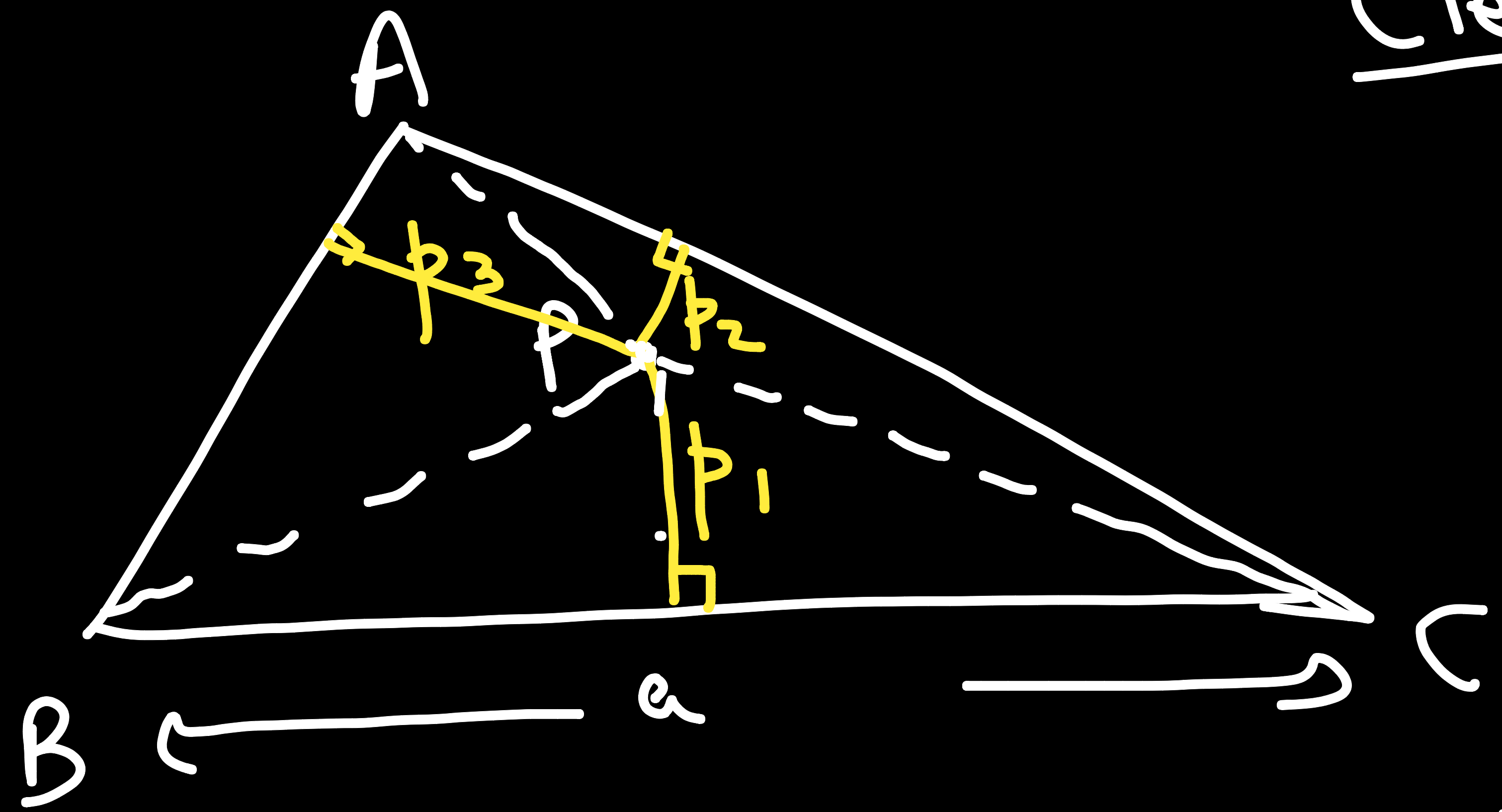
$$s - b = 6 \rightarrow a = 13$$

$$s - c = 7 \rightarrow b = 15$$

$$a = 13, b = 15, c = 14$$

Ques Let P lies inside the $\triangle ABC$, if p_1, p_2, p_3 are the length of perpendiculars from P to BC, CA and AB respectively, then find p_1, p_2, p_3 , such that $p_1^2 + p_2^2 + p_3^2$ is minimum.

Soln.



Clearly,

$$\Delta = \frac{1}{2} p_1 \cdot a + \frac{1}{2} p_2 \cdot b + \frac{1}{2} p_3 \cdot c$$

$$= \frac{a p_1 + b p_2 + c p_3}{2} \Rightarrow \sum a p_i = 2\Delta$$

Apply CS-Inequality

$$(a^2 + b^2 + c^2) \cdot (p_1^2 + p_2^2 + p_3^2) \geq (a p_1 + b p_2 + c p_3)^2$$

$$\Rightarrow p_1^2 + p_2^2 + p_3^2 \geq \frac{4\Delta^2}{\sum a^2}$$

Cauchy's Inequality (C.S-Inequality)

$$(a_1^2 + a_2^2 + \dots + a_n^2) \cdot (b_1^2 + b_2^2 + \dots + b_n^2) \geq (a_1 b_1 + a_2 b_2 + \dots + a_n b_n)^2$$

for equality

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = \dots$$

$$\Rightarrow p_1^2 + p_2^2 + p_3^2 \Big|_{\min} = \frac{4\Delta^2}{a^2 + b^2 + c^2}$$

for equality

$$\frac{p_1}{a} = \frac{p_2}{b} = \frac{p_3}{c} = \lambda \text{ (say)}$$

$$p_1 = a\lambda, \quad p_2 = b\lambda, \quad p_3 = c\lambda.$$

We also have

$$ap_1 + bp_2 + cp_3 = 2\Delta \Rightarrow a^2\lambda + b^2\lambda + c^2\lambda = 2\Delta$$
$$\lambda = \frac{2\Delta}{\sum a^2}.$$

Required values of p_1, p_2, p_3 are

$$p_1 = \frac{2a\Delta}{a^2+b^2+c^2}, \quad p_2 = \frac{2b\Delta}{a^2+b^2+c^2}$$

$$p_3 = \frac{2c\Delta}{a^2+b^2+c^2}.$$

Ans

$$\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$$

$$\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$$

$$\vec{a} \cdot \vec{b} = \sum a_i b_i = ab \cos(\theta)$$

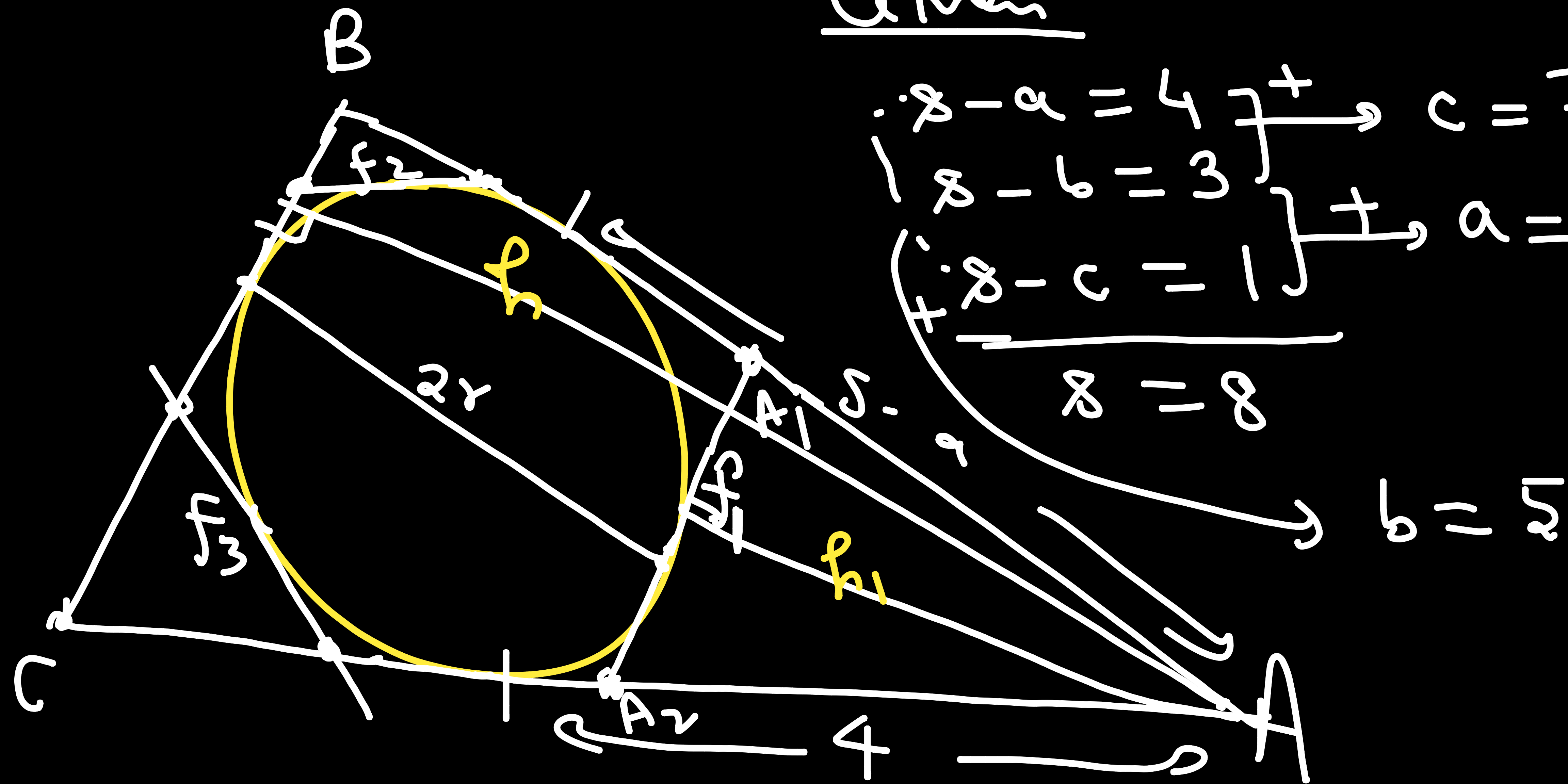
$\Downarrow$

$$(\sum a_i b_i)^2 = a^2 b^2 \cos^2(\theta) \leq a^2 b^2$$

$$(a_1 b_1 + a_2 b_2 + a_3 b_3)^2 \leq (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$$

Ques The length of the tangents drawn from A, B, C to the incircle of $\triangle ABC$ are 4, 3, 1 respectively. If lengths of part of the tangents within the triangle which are drawn parallel to the 3-sides BC, CA and AB of the $\triangle ABC$ are f_1, f_2, f_3 respectively then find $f_1 + f_2 + f_3$ to the circle.

Soln:



Given

$$\begin{aligned} s - a &= 4 \quad \text{---} \quad c = 7 \\ s - b &= 3 \quad \text{---} \quad a = 4 \\ s - c &= 1 \quad \text{---} \quad b = 5 \\ \hline s &= 8 \end{aligned}$$

$$b = 5$$

Clearly $\triangle AA_1A_2 \sim \triangle ABC$.

Hence $\frac{f_1}{BC} = \frac{r_1}{r} = \frac{r - 2r}{r} = 1 - \frac{2r}{r}$

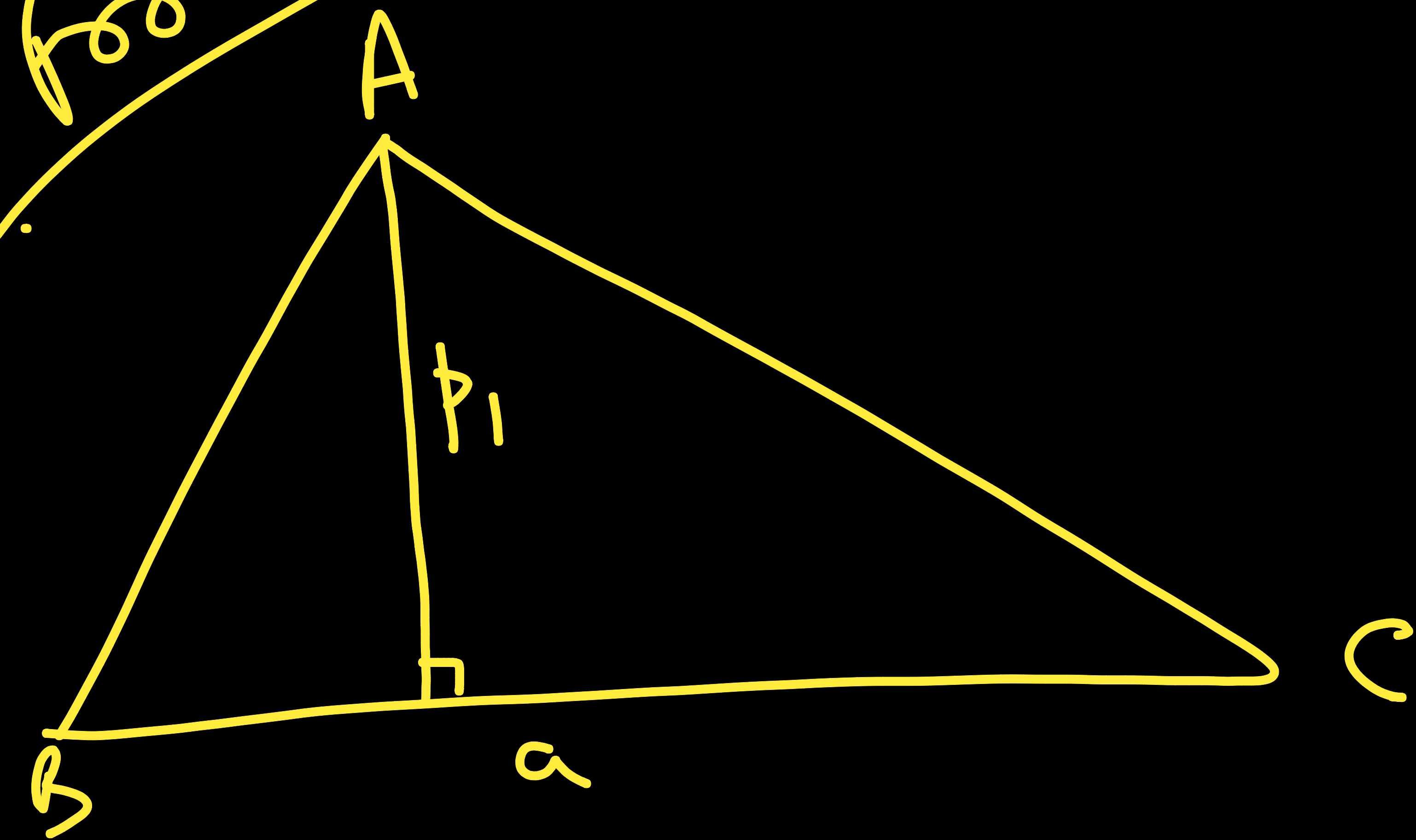
$$f_1 = a \cdot \left(1 - \frac{2r}{r}\right) = a \cdot \left(1 - \frac{2 \cdot \left(\frac{\Delta}{s}\right)}{\left(\frac{\Delta}{a}\right)}\right) = a \left(1 - \frac{a}{s}\right)$$

$$f_1 = \frac{a(s-a)}{s} = \frac{4 \times 4}{8} = 2$$

$$f_2 = \frac{b(s-b)}{s} = \frac{5 \times 3}{8} = \frac{15}{8}$$

$$f_3 = \frac{c(s-c)}{s} = \frac{7 \times 1}{8} = \frac{7}{8}$$

Formula



$$\Delta = \frac{1}{2} \cdot p_1 \cdot a$$

$$p_1 = \frac{2\Delta}{a}$$

$$f_1 = 2$$

$$f_2 = \frac{15}{8}$$

$$f_3 = \frac{7}{8}$$

$$\begin{aligned} & f_1 + f_2 - f_3 \\ &= 2 + \frac{15}{8} - \frac{7}{8} \\ &= \boxed{3} \text{ Ans} \end{aligned}$$

Ques find 'k' if $\sum_{n=1}^{10} \sum_{m=1}^{10} \tan^{-1}\left(\frac{m}{n}\right) = k \cdot \pi$.

Soln:

$$\begin{aligned} \text{L.H.S.} &= \sum_{n=1}^{10} \sum_{m=1}^{10} \tan^{-1}\left(\frac{m}{n}\right) \\ &= \left(\right) + \underbrace{\left(\tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{4}{3}\right) \right)}_{= \frac{\pi}{2}} + \dots + \left(\right) \\ &= 50 \cdot \frac{\pi}{2} \\ &= 25\pi \end{aligned}$$

$$\boxed{k=25}$$

Ques Find the integral values of K for which the system of equations $\cos^{-1}(x) + (\sin^{-1}(y))^2 = \frac{K\pi^2}{4}$ ① $(\sin^{-1}(y))^2 \cdot \cos^{-1}(x) = \frac{\pi^4}{16}$, has real solutions.

Soln:

Let $\cos^{-1}(x) = t$
 $t \in [0, \pi]$

$$y = 1$$

$$\begin{aligned} \frac{\pi^2}{4} + \frac{\pi^2}{4} &= \frac{2\pi^2}{4} \\ \cos^{-1}(x) + (\sin^{-1}(y))^2 &= \frac{K\pi^2}{4} \\ \cos^{-1}(x) \cdot (\sin^{-1}(y))^2 &= \frac{\pi^4}{16} \end{aligned}$$

$$t + (\sin^{-1}(y))^2 = \frac{K\pi^2}{4}$$

$$t \cdot (\sin^{-1}(y))^2 = \frac{\pi^4}{16}$$

$$t_{\min} = \frac{\left(\frac{\pi^4}{16}\right)}{\left(\frac{1}{2}\right)^2} = \frac{\pi^2}{4}$$

$$t \in \left[\frac{\pi^2}{4}, \pi\right]$$

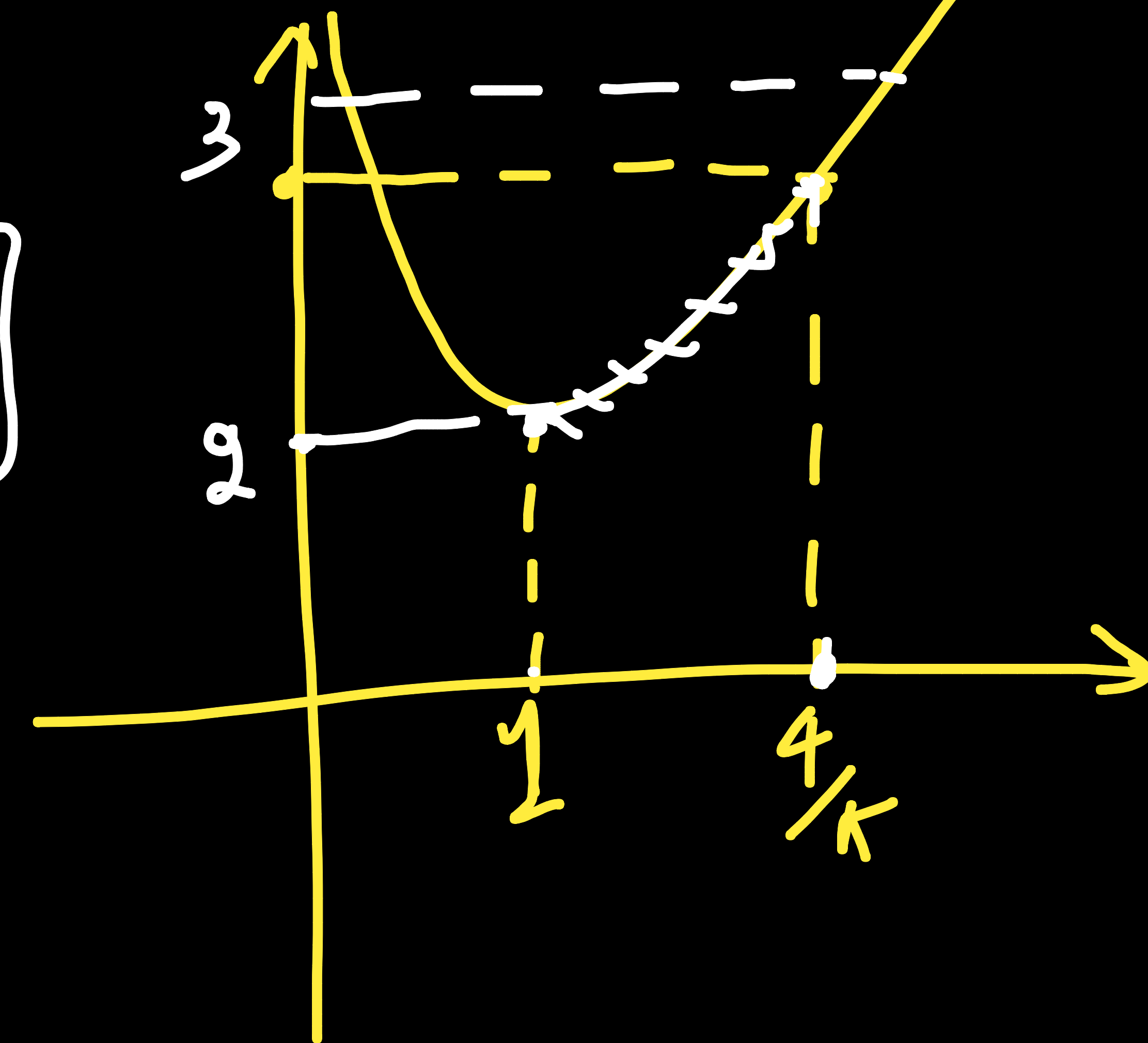
$$t + \frac{\left(\frac{\pi^4}{16}\right)}{t} = K \cdot \frac{\pi^2}{4}$$

$$\Rightarrow K = \boxed{\frac{4}{\pi^2} \cdot t} + \frac{\pi^2}{4t} = z + \frac{1}{z}$$

$$\left(z + \frac{1}{z}\right)_{z = \frac{4}{\pi}} = \frac{4}{\pi} + \frac{\pi}{4} = \frac{4}{3 \cdot 14} + \frac{3 \cdot 14}{4} = \underline{\underline{1 \cdot 3 + 8 \dots}}$$

$$z \in \left[1, \frac{4}{\pi}\right]$$

$$\boxed{K=2}$$



$$K=2$$

$$\cos^{-1}(x) = \frac{\pi^2}{4} \Rightarrow x = \cos\left(\frac{\pi^2}{4}\right)$$

$$\textcircled{2} \sin^{-1}(y) = \pm \frac{\pi}{2}$$

$$y = \pm 1$$

Ques Find positive integral solutions x and y of the equation $\rightarrow$

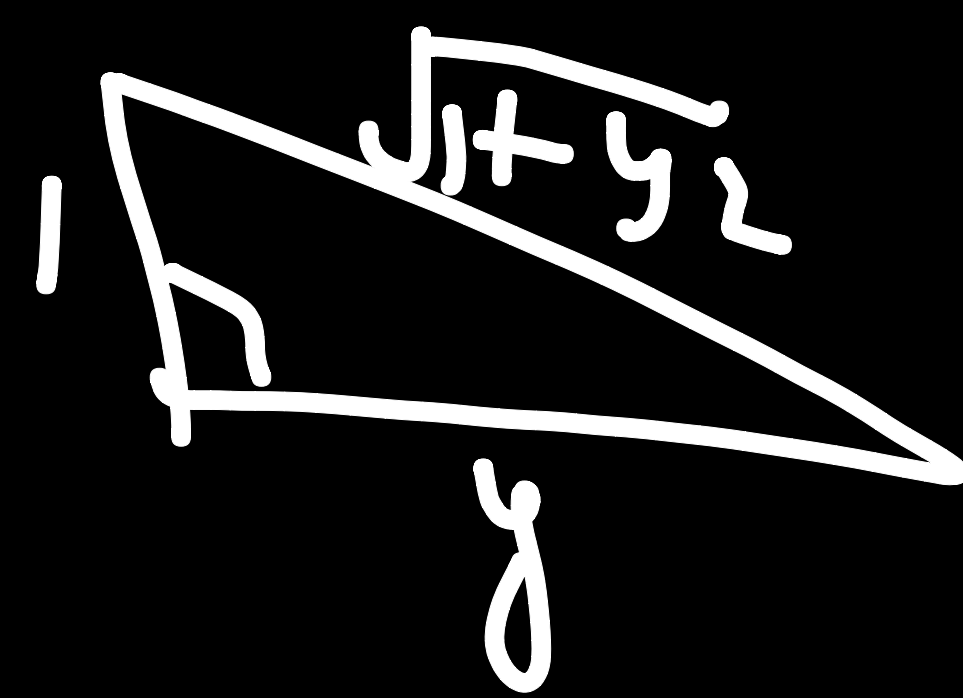
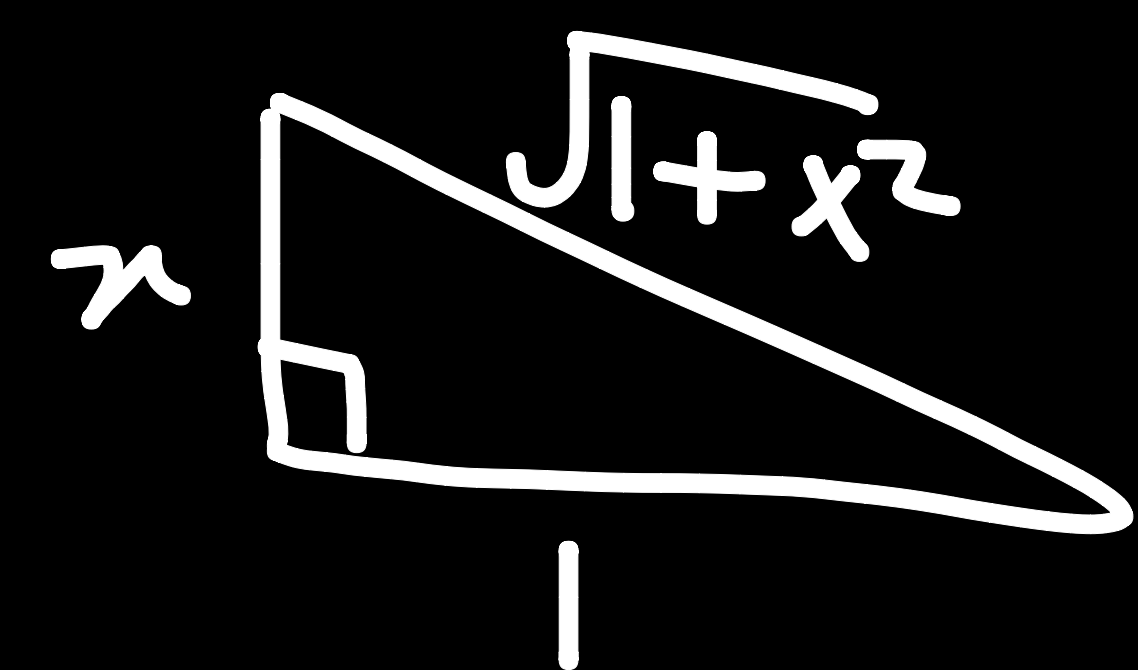
$$\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) + \cos^{-1}\left(\frac{y}{\sqrt{1+y^2}}\right) = \tan^{-1}(3).$$

Soln

$$\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) + \cos^{-1}\left(\frac{y}{\sqrt{1+y^2}}\right) = \tan^{-1}(3).$$

$$\tan^{-1}(x) + \tan^{-1}\left(\frac{1}{y}\right) = \tan^{-1}(3)$$

$$\tan^{-1}(x) = \tan^{-1}(3) - \tan^{-1}\left(\frac{1}{y}\right) = \tan^{-1}\left(\frac{3 - \frac{1}{y}}{1 + \frac{3}{y}}\right) = \tan^{-1}\left(\frac{3y-1}{y+3}\right)$$



$$x = \frac{3y-1}{y+3} = \frac{3(y+3) - 10}{y+3} = 3 - \frac{10}{y+3}$$

$$y+3 = 5, 10$$

$$y = 2, 7$$

$$y = 2 \Rightarrow x = 1$$

$$\textcircled{1} y = 7 \textcircled{2} x = 2$$

$$(x, y) = (2, 7) \textcircled{1} (1, 2)$$

Ans

Ques If a and b are positive quantities, such that.

$a_1 = \frac{a+b}{2}$, $b_1 = \sqrt{a_1 \cdot b}$, $a_2 = \frac{a_1 + b_1}{2}$, $b_2 = \sqrt{a_2 \cdot b_1}$ and so on...
 then find $\lim_{n \rightarrow \infty} a_n$ & $\lim_{n \rightarrow \infty} b_n$ in terms of a and b given ($b > a$).

Soln. Let $\underline{a} = b \cos(\theta) \Rightarrow \cos(\theta) = \frac{a}{b}$

$$\underline{a}_1 = \frac{a+b}{2} = \frac{b(1+\cos(\theta))}{2} = b \cos^2\left(\frac{\theta}{2}\right)$$

$$b_1 = \sqrt{a_1 \cdot b} = \sqrt{b \cos^2\left(\frac{\theta}{2}\right) \cdot b} = \underline{b \cos\left(\frac{\theta}{2}\right)}$$

$$a_2 = \frac{a_1 + b_1}{2} = \frac{b \cos^2\left(\frac{\theta}{2}\right) + b \cos\left(\frac{\theta}{2}\right)}{2}$$

$$= \underline{b \cos\left(\frac{\theta}{2}\right) \left(\cos\left(\frac{\theta}{2}\right) + 1\right)}$$

$$a_2 = \underline{b \cos\left(\frac{\theta}{2}\right) \cdot \cos\left(\frac{\theta}{4}\right)}$$

$$b_2 = \sqrt{a_2 \cdot b_1} = \sqrt{b \cos\left(\frac{\theta}{2}\right) \cos^2\left(\frac{\theta}{4}\right) \cdot b \cos\left(\frac{\theta}{2}\right)}$$

$$b_2 = b \cos\left(\frac{\theta}{2}\right) \cdot \underline{\cos\left(\frac{\theta}{4}\right)}$$

$$b_n = b \cdot \cos\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{4}\right) \cos\left(\frac{\theta}{8}\right) \dots \cos\left(\frac{\theta}{2^n}\right)$$

$$b_n = b \cdot \frac{\sin(\theta)}{2^n \cdot \sin\left(\frac{\theta}{2^n}\right)} \Rightarrow \lim_{n \rightarrow \infty} b_n = \frac{b \sin(\theta)}{\theta \cdot \lim_{n \rightarrow \infty} \frac{\sin\left(\frac{\theta}{2^n}\right)}{\left(\frac{\theta}{2^n}\right)}}$$

$$b_n = \frac{b \cdot \sin(\theta)}{\theta} = \frac{b \cdot \sqrt{1 - \frac{a^2}{b^2}}}{\cos^{-1}\left(\frac{a}{b}\right)}$$

$$b_n = \frac{\sqrt{b^2 - a^2}}{\cos^{-1}\left(\frac{a}{b}\right)} \cdot \underline{A}$$

$$a_1 = b \cos^2\left(\frac{\theta}{2}\right)$$

$$a_2 = b \cos\left(\frac{\theta}{2}\right) \cdot \cos^2\left(\frac{\theta}{4}\right)$$

$$a_3 = b \cos\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{4}\right) \cdot \cos^2\left(\frac{\theta}{2^3}\right)$$

⋮

$$a_n = b \cos\left(\frac{\theta}{2}\right) \cdot \cos\left(\frac{\theta}{4}\right) \cdots \cos\left(\frac{\theta}{2^{n-1}}\right) \cdot \cos^2\left(\frac{\theta}{2^n}\right)$$

$$a_n = b_n \cdot \cos\left(\frac{\theta}{2^n}\right)$$

$$n \rightarrow \infty \quad \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n \cdot \cos(0) = \lim_{n \rightarrow \infty} b_n = \frac{\sqrt{b^2 - a^2}}{\cos^{-1}\left(\frac{a}{b}\right)} \cdot \underline{A}$$